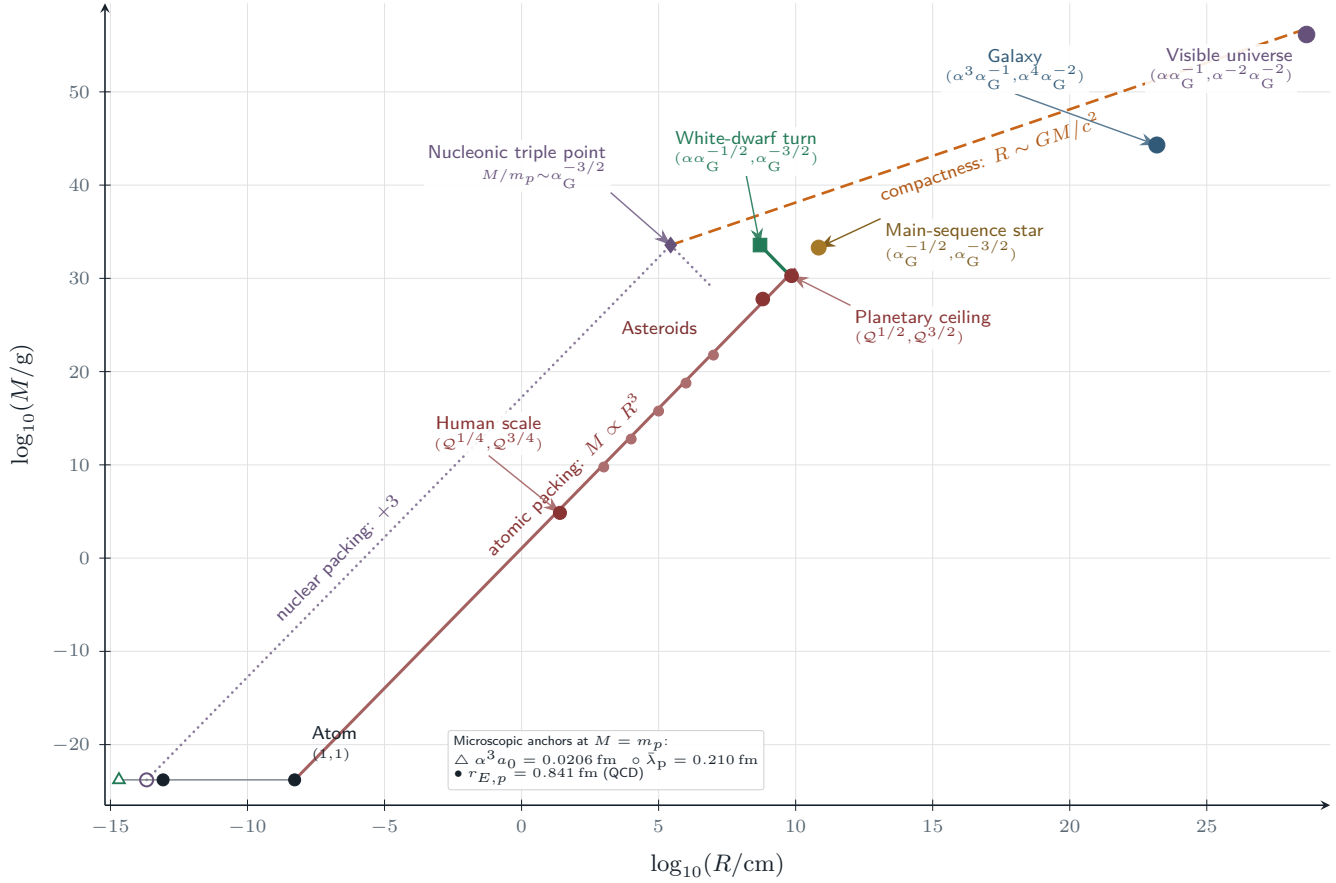


From α and α_G to the Architecture of Scale

Reading the mass–radius power graph

The graph below, from Barrow [1] compresses almost eighty orders of magnitude into two dimensionless couplings. Its exponents arise from a small collection of physical ideas: volume and area scaling, electromagnetic binding, gravitational self-energy, quantum pressure, radiative cooling and cosmic criticality. The estimates are coarse showing something structural: why atoms, organisms, planets, stars and galaxies occupy widely separated regions of the mass–radius plane.



The solid red line is exactly the atomic-packing law $M/m_p = (R/a_0)^3$ and is terminated at the Jupiter scale. The solid green line is the non-relativistic electron-degeneracy branch; the long-dashed orange line is the compactness slope $R \sim GM/c^2$. Violet dotted lines show the nucleonic analogues whose genuine triple point has mass $m_p \alpha_G^{-3/2}$. The Sun's proximity as a main-sequence star to an extrapolated R^3 line is an incidental equality of mean density; its equilibrium belongs to the stellar regime.

Power map at a glance. Each entry is a dimensionless location in the graph's natural units.

Scale	Radius R/a_0	Mass M/m_p
Atom	1	1
Human	$Q^{1/4}$	$Q^{3/4}$
Planetary ceiling	$Q^{1/2}$	$Q^{3/2}$
White-dwarf turn	$\alpha \alpha_G^{-1/2}$	$\alpha_G^{-3/2}$
Nucleonic triple	$\alpha \beta \alpha_G^{-1/2}$	$\alpha_G^{-3/2}$
Main-sequence star	$\alpha_G^{-1/2}$	$\alpha_G^{-3/2}$
Galaxy	$\alpha^3 \alpha_G^{-1}$	$\alpha^4 \alpha_G^{-2}$
Visible universe	$\alpha \alpha_G^{-1}$	$\alpha^{-2} \alpha_G^{-2}$

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} \simeq \frac{1}{137}, \quad \alpha_G = \frac{Gm_p^2}{\hbar c} \quad Q \equiv \frac{\alpha}{\alpha_G} = \frac{e^2/(4\pi\epsilon_0)}{Gm_p^2}$$

1. The number that opens the graph

The electromagnetic fine-structure constant is

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} \simeq \frac{1}{137}. \quad (1)$$

Using the proton mass as the gravitational reference gives

$$\alpha_G = \frac{Gm_p^2}{\hbar c} \simeq 5.9 \times 10^{-39} = \left(\frac{m_p}{m_{Pl}} \right)^2. \quad (2)$$

Their ratio is

$$\mathcal{Q} \equiv \frac{\alpha}{\alpha_G} = \frac{e^2/(4\pi\epsilon_0)}{Gm_p^2} \simeq 1.2 \times 10^{36}. \quad (3)$$

It is the ratio of electrical repulsion to gravitational attraction between two protons. The immense separation of microscopic and astronomical scales is already present in this single number:

$$\mathcal{Q}^{1/4} \sim 10^9, \quad \mathcal{Q}^{3/2} \sim 10^{54}.$$

The graph measures masses in proton masses and radii in atomic radii:

$$M = m_p \mathcal{M}, \quad R = a_0 \mathcal{R}, \quad (4)$$

where

$$a_0 = \frac{\hbar}{\alpha m_e c} \quad (5)$$

is the Bohr radius.

2. Compton length per Bohr radius lengths

The claim that only α and α_G appear uses one additional empirical compression:

$$\beta \equiv \frac{m_e}{m_p} \sim \alpha^2. \quad (6)$$

Numerically, $\beta = 5.45 \times 10^{-4}$ while $\alpha^2 = 5.33 \times 10^{-5}$, so the approximation is accurate only to one order of magnitude. It converts the reduced proton Compton length $\bar{\lambda}_p = \hbar/(m_p c)$ into

$$\frac{\bar{\lambda}_p}{a_0} = \alpha\beta \sim \alpha^3. \quad (7)$$

This dimensional compression supplies a third length, $\alpha^3 a_0$. It must be kept separate from both the measured charge radius and the Compton length. Using current recommended constants gives [7]

$$r_{E,p} = 0.841 \text{ fm}, \quad \log_{10} \left(\frac{a_0}{r_{E,p}} \right) = 4.80, \quad (8)$$

$$\bar{\lambda}_p = 0.210 \text{ fm}, \quad \log_{10} \left(\frac{a_0}{\bar{\lambda}_p} \right) = 5.40, \quad (9)$$

$$\alpha^3 a_0 = 0.0206 \text{ fm}, \quad \log_{10} \left(\frac{a_0}{\alpha^3 a_0} \right) = 6.41. \quad (10)$$

The filled proton marker on the graph is therefore the QCD-set charge radius. The open circle marks $\bar{\lambda}_p$, and the triangle marks the two-constant $\alpha^3 a_0$ estimate. No single “proton radius” represents all three.

3. Atomic density and the cubic line

Condensed matter contains roughly one proton mass in each atomic-sized volume:

$$\rho_0 \sim \frac{m_p}{a_0^3}. \quad (11)$$

An object containing N atoms consequently has

$$M \sim N m_p, \quad R \sim a_0 N^{1/3}. \quad (12)$$

Eliminating N gives

$$\boxed{\frac{M}{m_p} \sim \left(\frac{R}{a_0} \right)^3}. \quad (13)$$

Writing $x = \log_{10}(R/\text{cm})$ and $y = \log_{10}(M/\text{g})$ turns this exact plotted scaling into

$$y = 3x + \log_{10} \left(\frac{m_p}{a_0^3} \right) \simeq 3x + 1.05. \quad (14)$$

A spherical body with mean density 1.4 g cm^{-3} instead obeys $y = 3x + \log_{10}(4\pi\rho/3) \simeq 3x + 0.77$. The two lines are close and remain distinct.

This is the diagonal sequence through organisms, asteroids and planets. Its slope is three on a logarithmic mass-radius diagram. The triple exponents in the mass formulae often arise from this geometrical fact. The plotted branch ends at the Jupiter scale. The Sun happens to have mean density 1.41 g cm^{-3} , so an extrapolated constant-density line passes near it on a logarithmic plot. Stellar hydrostatic equilibrium follows different physics, and the proximity has no scaling significance.

4. The planetary ceiling

For a constant-density body, Eq. (13) gives

$$M \sim m_p \left(\frac{R}{a_0} \right)^3. \quad (15)$$

The gravitational binding energy of a surface atom is then

$$E_G \sim \frac{GMm_p}{R} \sim \frac{Gm_p^2}{a_0} \left(\frac{R}{a_0} \right)^2. \quad (16)$$

The electromagnetic binding scale is

$$E_{em} \sim \frac{e^2}{4\pi\epsilon_0 a_0}. \quad (17)$$

When $E_G \sim E_{em}$, self-gravity begins to compress and ionise ordinary matter. Hence

$$\left(\frac{R_P}{a_0} \right)^2 \sim \frac{\alpha}{\alpha_G} = \mathcal{Q}, \quad (18)$$

and therefore

$$\boxed{\frac{R_P}{a_0} \sim \mathcal{Q}^{1/2}, \quad \frac{M_P}{m_p} \sim \mathcal{Q}^{3/2}}. \quad (19)$$

The estimates give $R_P \sim 6 \times 10^9 \text{ cm}$ and $M_P \sim 2 \times 10^{30} \text{ g}$, close to Jupiter. This is an upper planetary scale rather than a prediction for every planet.

5. Why the organism scale is a quarter power

The traditional “Man” estimate combines complex chemistry, a planet capable of retaining a processed atmosphere, and an organism large enough for a fall to threaten structural failure. Let the organism have size L and atomic density. Then

$$M_L \sim m_p \left(\frac{L}{a_0} \right)^3. \quad (20)$$

Its fall energy scales as volume times height:

$$E_{\text{fall}} \sim M_L g L \propto L^4. \quad (21)$$

A fracture crosses an area containing approximately $(L/a_0)^2$ bonds, so

$$E_{\text{break}} \sim E_{\text{em}} \left(\frac{L}{a_0} \right)^2 \propto L^2. \quad (22)$$

Equating the two and inserting the characteristic planetary gravity obtained above yields

$$\boxed{\frac{L}{a_0} \sim \mathcal{Q}^{1/4}, \quad \frac{M_L}{m_p} \sim \mathcal{Q}^{3/4}.} \quad (23)$$

The quarter power is a square root nested inside the planetary square root. Press’s fuller estimate introduces a chemical factor $\epsilon^{1/4}$ and biomechanical coefficients. The powers identify a scale; they do not fix a human body with biological precision.

6. Stellar mass and the phase geometry

Consider N baryons in a self-gravitating object. Its gravitational energy is

$$E_G \sim \frac{G m_p^2 N^2}{R}. \quad (24)$$

Relativistic quantum support has the characteristic scale

$$E_Q \sim \frac{\hbar c N^{4/3}}{R}. \quad (25)$$

Equating these energies cancels R and gives

$$N^{2/3} \sim \frac{1}{\alpha_G}, \quad N \sim \alpha_G^{-3/2}. \quad (26)$$

Thus

$$\boxed{M_{\text{Ch}} \sim m_p \alpha_G^{-3/2} = \frac{m_{\text{Pl}}^3}{m_p^2}.} \quad (27)$$

This is the Chandrasekhar mass scale, approximately a solar mass [2]. It reappears throughout stellar physics even though an ordinary main-sequence star is supported thermally. A separate mean-density estimate gives

$$\frac{R_\star}{a_0} \sim N^{1/3} \sim \alpha_G^{-1/2}, \quad (28)$$

which is close to the solar radius.

The same limiting mass can be reached geometrically. Before the electrons become relativistic, balancing their degeneracy energy against gravity gives

$$\frac{R}{a_0} \sim \mathcal{Q} N^{-1/3}, \quad \frac{M}{m_p} = N, \quad (29)$$

and hence the descending branch

$$\boxed{\frac{M}{m_p} \sim \mathcal{Q}^3 \left(\frac{R}{a_0} \right)^{-3}.} \quad (30)$$

This is the -3 slope of non-relativistic white dwarfs [3]. It meets the atomic $+3$ branch at the Jupiter scale. The electron Fermi momentum becomes relativistic when

$$p_F \sim \frac{\hbar N^{1/3}}{R} \sim m_e c, \quad \frac{R}{a_0} \sim \alpha N^{1/3}. \quad (31)$$

Combining this boundary with Eq. (30) yields

$$N \sim \alpha_G^{-3/2}, \quad \frac{R_{\text{WD}}}{a_0} \sim \alpha \alpha_G^{-1/2}. \quad (32)$$

The white-dwarf branch therefore reaches the mass in Eq. (27) at a radius of order 5×10^8 cm.

The compactness line has slope $+1$. Suppressing the horizon’s factor of two,

$$R_g \sim \frac{GM}{c^2}. \quad (33)$$

Its clean triple point appears after changing from the atomic ruler a_0 to the nucleonic ruler $\bar{\lambda}_p$. Put $r = R/\bar{\lambda}_p$. Nuclear packing, non-relativistic neutron degeneracy, and compactness then read

$$N \sim r^3, \quad \text{nuclear packing (+3),} \quad (34)$$

$$N \sim \alpha_G^{-3} r^{-3}, \quad \text{neutron degeneracy (-3),} \quad (35)$$

$$N \sim \alpha_G^{-1} r, \quad \text{compactness (+1).} \quad (36)$$

All three meet at

$$\boxed{r \sim \alpha_G^{-1/2}, \quad N \sim \alpha_G^{-3/2}.} \quad (37)$$

This nucleonic triple point has the Chandrasekhar mass ordinate and a radius of order $GM_{\text{Ch}}/c^2 \sim 3 \times 10^5$ cm. The electron-degenerate endpoint lies farther right by

$$\frac{R_{\text{WD}}}{R_g} \sim \frac{1}{\beta} \sim 10^3. \quad (38)$$

The white-dwarf line and the compactness line consequently have separate endpoints. Their shared mass ordinate is the structural result: once quantum support is relativistic, the fermion rest mass drops out and α_G alone fixes the limiting mass.

7. Galaxies: collapse must be able to cool

A protogalactic cloud forms stars only if radiation removes its thermal energy on roughly the gravitational collapse time:

$$t_{\text{cool}} \sim t_{\text{ff}}. \quad (39)$$

Atomic recombination and bremsstrahlung introduce α and m_e . Free fall introduces G . Combining this condition with the Jeans scale gives

$$\frac{M_{\text{gal}}}{m_p} \sim \alpha^5 \alpha_G^{-2} \beta^{-1/2}, \quad (40)$$

$$\frac{R_{\text{gal}}}{a_0} \sim \alpha^4 \alpha_G^{-1} \beta^{-1/2}. \quad (41)$$

Using the compression $\beta \sim \alpha^2$ from Eq. (6) gives the powers printed on the graph:

$$\boxed{\frac{M_{\text{gal}}}{m_p} \sim \alpha^4 \alpha_G^{-2}, \quad \frac{R_{\text{gal}}}{a_0} \sim \alpha^3 \alpha_G^{-1}.} \quad (42)$$

These correspond to about 7×10^{10} solar masses and 10^2 kpc. Dark matter, feedback, chemical composition and formation history broaden the actual galactic population.

8. The visible-universe point

At the top of the graph,

$$R_U \sim a_0 \frac{\alpha}{\alpha_G}, \quad M_U \sim m_p \alpha^{-2} \alpha_G^{-2}. \quad (43)$$

Within the approximations already used, these satisfy

$$\boxed{\frac{GM_U}{R_U c^2} \sim 1.} \quad (44)$$

The same order-of-magnitude relation holds for the mass contained within a Hubble radius in a universe near critical density. It does not identify the universe with an ordinary black hole. The radius estimate also incorporates an epoch-selection argument: observers arise while long-lived stars are available.

This last point is more contingent than the atomic, planetary and stellar scalings. Contemporary cosmology needs the Hubble parameter, matter density, dark energy and primordial conditions. The two couplings alone cannot specify the present observable universe.

9. Summarising the graph

The hierarchy follows a short logical chain:

- α establishes atomic size and binding energy.
- α_G establishes the weakness of gravity per proton.
- Packing gives slope +3, non-relativistic degeneracy gives −3, and relativistic compactness gives +1.
- Balance conditions introduce square roots and fractional powers.
- Enormous particle numbers allow weak gravity to become decisive.

The diagram is a map of dominant balances and their domains of validity. Its two −3 branches also expose the distinction between a repeated slope and a repeated physical object: electron degeneracy describes white dwarfs, while the nucleonic analogue completes the compactness

triple point. The omitted numerical coefficients and additional dimensionless parameters matter greatly for any specific object. The persistence of the hierarchy after those omissions is the significant result.

Two continuations of the scale map

The organism point admits a chemical and thermodynamic refinement as Press's full estimate introduces the weak-molecular parameter ϵ , while Kleiber scaling concerns the subsequent transport of metabolic power through organised living matter; (See [From Schrödinger to Press to Kleiber](#)).

At the opposite end of the diagram, the compactness relation $GM_U/(R_U c^2) \sim 1$ leads to the horizon area law,

$$\frac{S_U}{k_B} \sim \pi \left(\frac{R_U}{\ell_P} \right)^2 \sim \pi \beta^{-2} \alpha_G^{-3}.$$

(See [Horizon Entropy and the Area Law](#)).

References

- [1] J. D. Barrow, *Theories of Everything: The Quest for Ultimate Explanation*, Oxford University Press, 1991.
- [2] S. Chandrasekhar, "The highly collapsed configurations of a stellar mass," *Monthly Notices of the Royal Astronomical Society* **91**, 456–466 (1931). doi:10.1093/mnras/91.5.456.
- [3] M. Nauenberg, "Analytic approximations to the mass-radius relation and energy of zero-temperature stars," *The Astrophysical Journal* **175**, 417–430 (1972). doi:10.1086/151568.
- [4] W. H. Press, "Man's size in terms of fundamental constants," *American Journal of Physics* **48**, 597–598 (1980). doi:10.1119/1.12326.
- [5] W. H. Press and A. P. Lightman, "Dependence of macrophysical phenomena on the values of the fundamental constants," *Philosophical Transactions of the Royal Society A* **310**, 323–336 (1983). doi:10.1098/rsta.1983.0094.
- [6] A. Burrows and J. P. Ostriker, "The astronomical reach of fundamental physics," *Proceedings of the National Academy of Sciences* **111**, 2409–2416 (2014). doi:10.1073/pnas.1318003111.
- [7] E. Tiesinga, P. J. Mohr, D. B. Newell and B. N. Taylor, "The 2022 CODATA recommended values of the fundamental physical constants," National Institute of Standards and Technology, version 9.0 (2024), <https://physics.nist.gov/constants>.